< Concept Paper>

Title:

Fintech AI Governance: Engineering a 'Human-AI Symbiosis' Protocol for System Reliability and Full-Spectrum Adoption

Research Background: The Engineering Challenge of Probabilistic Systems

Most current industrial applications treat Generative AI (GenAI) as a simple efficiency tool. However, enterprise-level deployment — such as the "Chat-to-Work" infrastructure applied to large-scale operations—reveals a structural engineering challenge. Traditional Fintech service systems are built on "Deterministic SOPs" (Standard Operating Procedures). The introduction of GenAI agents forces a shift toward "Probabilistic Workflows," where the system's output reliability heavily depends on human-AI collaboration rather than rigid code execution.

The Problem: While the technical infrastructure is established, there is a significant "System Performance Gap" during enterprise adoption. Variance in operational efficiency and error rates cannot be explained by algorithms alone. From a Management Systems Engineering perspective, the missing link is a robust governance protocol that aligns user acceptance (micro-level) with organizational deployment strategies (macro-level).

Theoretical Framework: Integrating TOE and TAM for AI Governance To model the mechanics of "Human-AI Symbiosis," this study proposes a comprehensive evaluation framework integrating the Technology-Organization-Environment (TOE) framework with the Technology Acceptance Model (TAM). This models how macro-level governance dictates micro-level trust and subsequent system reliability.

- **The Technology-Environment Context (TOE & System Parameters):**
    - **Technological Context:** System Explainability & Output Consistency. How do the inherent probabilistic risks of GenAI impact the perceived baseline reliability of the system?
    - **Organizational Context:** Management Systems Agility. Is the operational hierarchy engineered to support decentralized Human-AI command loops?
    - **Environmental Context:** Fintech Regulatory Governance. How do strict

compliance boundaries constrain or shape the system's operational design?

- **The Adoption Mechanism (TAM Integration):**
  - **Perceived System Trust & Usefulness:** Bridging TOE and TAM, this study posits that "Trust-in-Process" acts as the critical mediator. Do employees perceive the probabilistic AI as a reliable co-pilot (Perceived Usefulness) without causing critical cognitive overload (Perceived Ease of Use)?
- **The Outcome (Dependent Variables):**
  - **Full-Spectrum Adoption & System Reliability:** Evaluated through measurable Industrial Engineering metrics: System Throughput Efficiency (Time-to-Decision) and Operational Quality (Error Reduction Rate under human-AI collaborative loops).
- **Methodology: A Data-Driven "Living Lab" Approach**

To ensure rigorous, engineering-grade validation suitable for an ISE Ph.D. dissertation, this study will employ Quantitative PLS-SEM (Partial Least Squares Structural Equation Modeling), triangulating dual-source data from a real-world Fintech deployment:

  - **Perceptual Data (TAM/TOE Assessment):** Surveying 200+ operational and risk management staff to quantify Trust, Perceived Usefulness, and Organizational Agility indices.
  - **Objective Engineering Data (System Logs):** Extracting hard operational metrics from the backend (e.g., measuring the delta in "Task Completion Time" and "Exception Handling Rates" before and after the "Chat-to-Work" transition).

Significance & Contribution

This research contributes directly to Management Systems Engineering by providing an empirical blueprint for "Fintech AI Governance." By proving how TOE and TAM factors interact to build Trust-in-Process, this study delivers a scalable 'Human-AI Symbiosis' protocol. It equips enterprises with the engineering management tools necessary to safely deploy probabilistic GenAI systems within high-stakes, highly regulated financial environments.